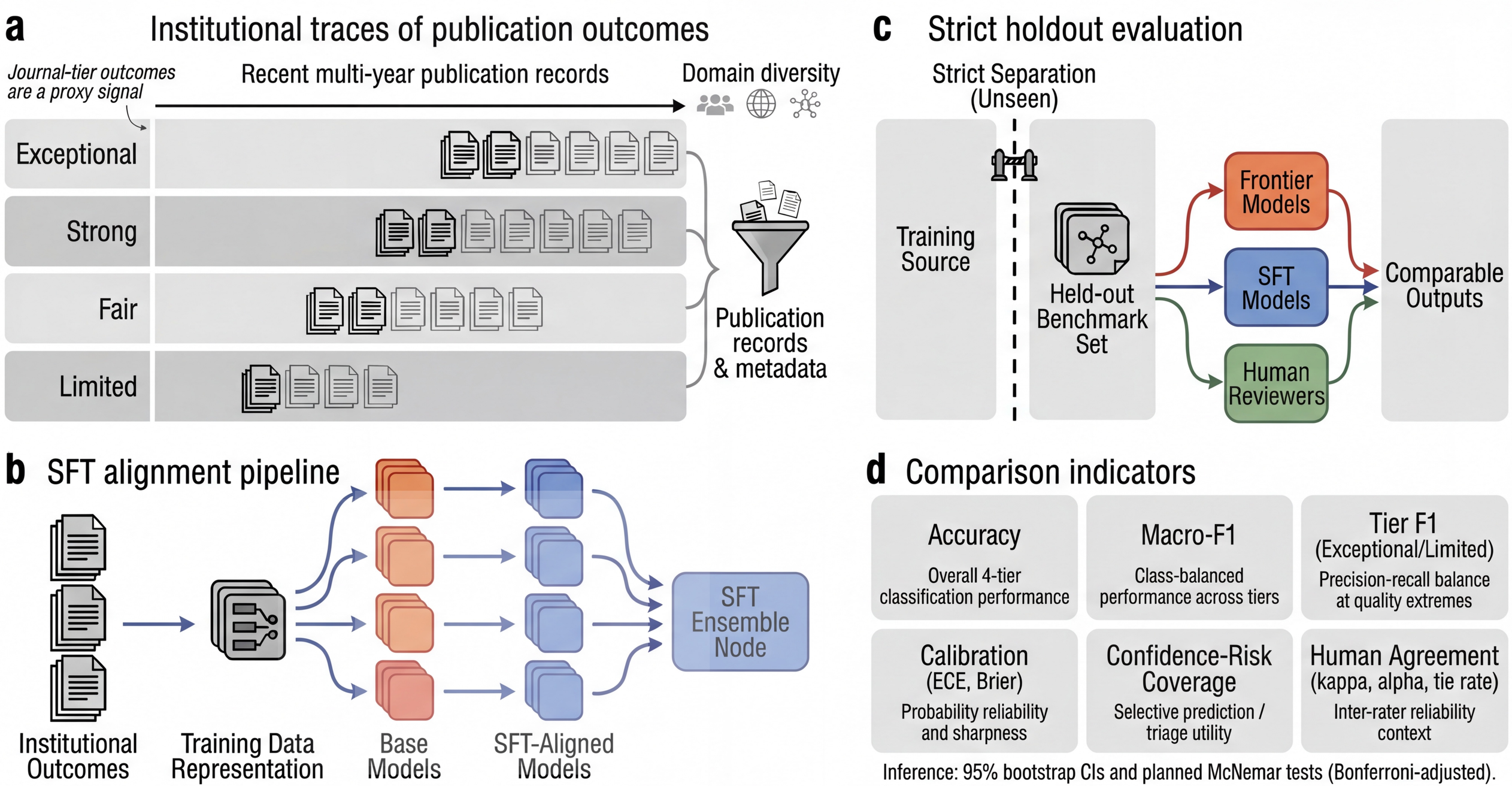
a
Institutional traces of publication outcomes
Journal-tier outcomes are a proxy signal
Recent multi-year publication records
Domain diversity
Exceptional
Strong
Fair
Limited
Publication records & metadata
b
SFT alignment pipeline
SFT Ensemble Node
Institutional Outcomes
Training Data Representation
Base Models
SFT-Aligned Models
c
Strict holdout evaluation
Strict Separation (Unseen)
Training Source
Held-out Benchmark Set
Frontier Models
SFT Models
Human Reviewers
Comparable Outputs
d
Comparison indicators
Accuracy
Overall 4-tier classification performance
Macro-F1
Class-balanced performance across tiers
Tier F1 (Exceptional/Limited)
Precision-recall balance at quality extremes
Calibration (ECE, Brier)
Probability reliability and sharpness
Confidence-Risk Coverage
Selective prediction / triage utility
Human Agreement (kappa, alpha, tie rate)
Inter-rater reliability context
Inference: 95% bootstrap CIs and planned McNemar tests (Bonferroni-adjusted).